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PROTEUS: simulating the coupled interior-atmosphere evolution of rocky planets through deep time

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Summary

PROTEUS (proteus-framework.org) is an open-source framework that simulates the coupled evolution of the interiors and atmospheres of rocky planets and sub-Neptunes through deep time. A simulation begins in the magma ocean epoch of a young planet and ends with a solid mantle or a permanently molten interior, beneath an atmosphere that is outgassed, altered, or lost. Interior structure, mantle energetics, volatile exchange, atmospheric energy balance, escape, the host star, tides, and accretion events are each modelled by a separate module. PROTEUS integrates these modules forward in time, so that each responds to the current state of the others, and derives synthetic spectra and phase curves from the evolved planet. Because the young terrestrial planets of the Solar System passed through the same regime, the framework also serves to reconstruct the early Earth, Venus, Mars, and Mercury. PROTEUS is written in Python and configured through a single TOML file. Every change to the framework or its modules has to pass automated unit tests on GitHub Actions before being merged, and integration and physics tests run every night.

Statement of need

Telescopes such as JWST now measure the atmospheres of super-Earths and sub-Neptunes (Kreidberg & Stevenson, 2025; Lichtenberg et al., 2025). However, mass and radius alone do

not determine the interior structure and composition, because one bulk density is consistent with many combinations of core, mantle, and envelope. Structure models used to address this degeneracy typically assume a cold, solid mantle, although every rocky planet begins its evolution molten (Lichtenberg & Miguel, 2025). Many observed low-mass planets are irradiated strongly enough to stay partially molten for their entire lifetime (Calder et al., 2026; Nicholls, Lichtenberg, Chatterjee, et al., 2026). A solid interior may therefore be the exception among the planets observed so far.

The state of such a planet emerges from the interplay between interior and atmosphere, and it cannot be computed one component at a time. Outgassed volatiles set the greenhouse forcing and hence the surface temperature, which governs the melt fraction of the mantle (Lichtenberg et al., 2021; Nicholls et al., 2024). Melt fraction and magma redox state in turn control how much of each volatile stays dissolved (Sastre, Lichtenberg, Soucasse, et al., 2026). Stellar evolution and atmospheric escape remove volatiles from this cycle (Cesario et al., 2026; Postolec, Lichtenberg, Nicholls, et al., 2026). Tides heat the mantle (Nicholls, Guimond, et al., 2025; van Dijk et al., 2026), while deep radiative layers in the atmosphere can slow its cooling (Nicholls, Pierrehumbert, Lichtenberg, Soucasse, et al., 2025). A cooling planet also does not reach the same climate as a planet that warms (Boer et al., 2025), so its present state cannot be derived from its current conditions alone. It has to be obtained by integrating the coupled system forward in time.

These feedbacks depend on laboratory measurements of material properties, such as equations of state (Attia et al., 2026), volatile solubilities (Sossi et al., 2023; Suer et al., 2023), and gas opacities. Laboratory experiments increasingly probe the pressures, temperatures, and compositions expected in exoplanet interiors, many of which have no equivalent in the Solar System (Lichtenberg & Miguel, 2025; Thompson, 2026). An evolutionary model therefore needs to remain adaptable, so that new empirical measurements can be included for arbitrary thermodynamic states and compositions as they become available.

The rocky planets of the inner Solar System passed through the same magma ocean epoch, but their later evolution has largely erased its record (Elkins-Tanton, 2012; Lichtenberg et al., 2023). Strongly irradiated exoplanets, in contrast, may reside in this phase today, which opens the magma ocean epoch to direct observation. With PROTEUS we have modelled the contrasting early evolution of Earth and Venus (Nicholls, Krissansen-Totton, et al., 2026) and the cooling of the Hadean Earth to clement surface conditions (van Dijk et al., 2026). The interior compositions and geochemistries it covers (Lichtenberg, Attia, et al., 2026; Nicholls et al., 2024) range from the reduced mantle of Mercury to the oxidised mantle of Mars. For exoplanets, the simulations predict the atmospheric composition, radius, and thermal emission that JWST and the observatories that follow it can measure. A measured spectrum or bulk density can thus be interpreted with PROTEUS as a snapshot of the evolutionary history of a planet. PROTEUS therefore connects laboratory measurements, the geological record of the Solar System, and telescope observations of exoplanets within one evolutionary model.

State of the field

Atmosphere models frequently neglect the dissolution of volatiles into the magma (Selsis et al., 2023; Turbet et al., 2021). Magma ocean models, conversely, often omit the greenhouse effect of their own atmosphere (Herath et al., 2024; Monteux et al., 2016; Solomatov, 2015). Coupled models of a magma ocean and its atmosphere date back to Abe & Matsui (1985) and Abe (1993), and were extended by many later studies (Elkins-Tanton, 2008; Hamano et al., 2013; Lebrun et al., 2013; Salvador et al., 2017; Schaefer et al., 2016). Later codes included atmospheric escape, geochemistry, or tidal heating (Barnes et al., 2020; Bower et al., 2022; Kite & Barnett, 2020; Krissansen-Totton et al., 2021; Lichtenberg et al., 2021). Recent studies extend such models to super-Earths, sub-Neptunes, and lava worlds (Calder et al., 2026; Carone et al., 2025; Cherubim et al., 2025; Farhat et al., 2025; Maurice et al., 2024; Nicholls, Lichtenberg, Chatterjee, et al., 2026; Sahu et al., 2025; Tang et al., 2025). However,

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not all of these codes are public, and each fixes one set of physical prescriptions in one code base.

We initiated the CHILI intercomparison to test these codes on common Earth, Venus, and exoplanet cases (Lichtenberg, Schaefer, et al., 2026; Nicholls, Krissansen-Totton, et al., 2026). For Earth the models agree on the solidification time to within 4 Myr, whereas for Venus they diverge and their outgassed atmospheres differ in surface pressure and dominant species. Two of the CHILI codes show how the underlying choices differ in practice. GOOEY (Schaefer et al., 2016) evolves a steam atmosphere of pure H_2O with pre-computed line-by-line cooling rates, and it tracks how the oxygen left behind by fractionating escape oxidises the magma ocean. PACMAN (Krissansen-Totton et al., 2021, 2024) extends this approach to C-H-O atmospheres with graphite, traps volatiles in the solidifying mantle, and evolves the mantle redox state together with the magma ocean. PROTEUS, in contrast, resolves the melt fraction through the depth of the mantle and solves for radiative-convective equilibrium in the atmosphere, so that deep radiative layers can form. The model spread originates from the treatment of mantle dynamics (boundary-layer scaling against a resolved 1-D mantle) and of radiative transfer (grey against non-grey, with or without deep radiative layers). Volatile trapping in the solid mantle and atmospheric escape add to it, and together these choices outweigh the sensitivity to the initial volatile inventory. Hence, the choice of model can bias the interpretation of observations. Such disagreement can be traced to a specific process only if components can be exchanged within otherwise identical simulations, which motivates the modular design of PROTEUS.

VPLanet (Barnes et al., 2020) is open source and similarly modular, but it describes the magma ocean and the atmosphere with parameterised box models. Magrathea (Huang et al., 2022) computes static interior structures, and MESA (Paxton et al., 2011) evolves stars and giant planets, but neither code follows the coupled evolution of a mantle and its atmosphere. To our knowledge, no other open-source framework resolves the planet in one dimension, from the core to the top of the atmosphere. The melt fraction is tracked radially through a mantle that crystallises and contracts (Lichtenberg, Attia, et al., 2026). The synthetic observables can also be fitted to observations by Bayesian optimisation of the evolutionary parameters (Nicholls, Lichtenberg, Riegler, et al., 2026).

Software design

PROTEUS keeps the coupling between components separate from their physics, and each module resides in its own repository with its own tests and documentation. The framework itself contains the coupling loop, the configuration schema, the input and output layer, and tools for exploring parameter grids on computer clusters (Figure 1).

In each iteration, the mantle module returns the temperature profile and melt fraction for the current atmospheric boundary condition, and the structure module updates radius and gravity. The accretion, tides, and star modules then update planet mass, tidal heating, and irradiation, and escape removes mass under the stellar spectrum. The outgassing module partitions volatiles between melt and atmosphere at the current melt mass and oxygen fugacity. Finally, the atmosphere module solves for radiative-convective equilibrium with the new composition. The time step adapts to the most rapidly changing quantity.

This operator-split approach results in a more complex installation process, more dependencies, and data exchange across Python, Julia, Fortran, and C code. However, it allows modules to be developed and verified independently, both standalone and within the coupled system, and existing codes to be integrated rather than re-implemented. Because modules with the same role are interchangeable, the sensitivity to a modelling choice can be measured within one framework. Most roles also have a simplified "dummy" module that runs in seconds, which keeps the coupled test suite fast and isolates the effect of a single component.

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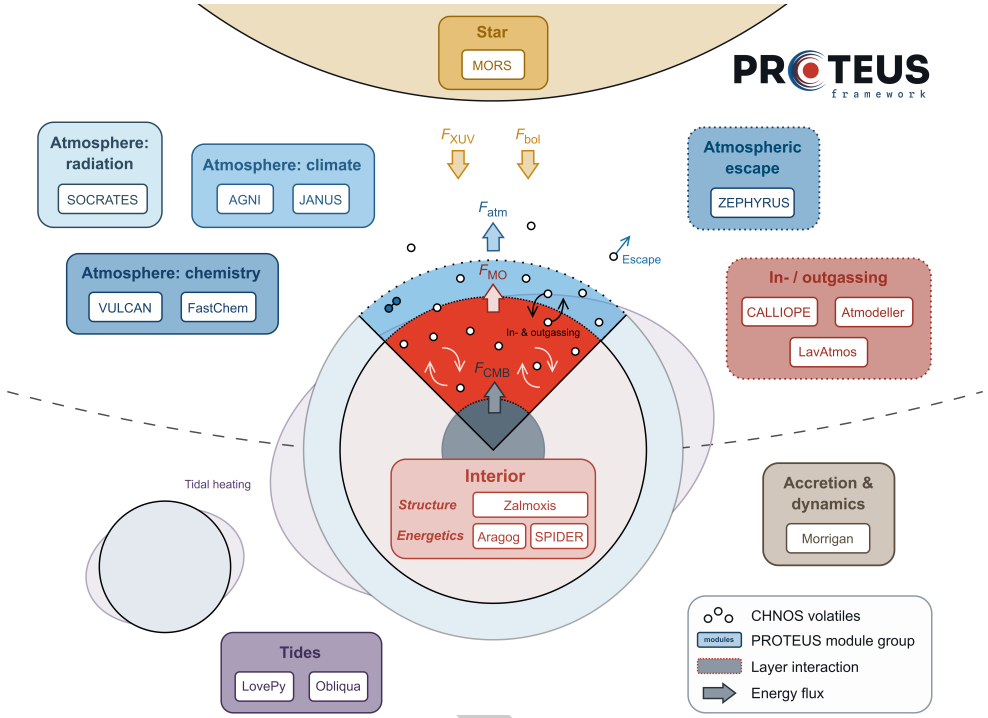


Figure 1: Module groups of the PROTEUS framework. Interior structure: Zalmoxis (Lichtenberg, Attia, et al., 2026), on the PALEOS equations of state (Attia et al., 2026). Mantle energetics: Aragog (Lichtenberg, Attia, et al., 2026), SPIDER (Bower et al., 2018), and a boundary-layer model. Atmosphere: AGNI (Nicholls, Pierrehumbert, Lichtenberg, Soucasse, et al., 2025; Nicholls, Pierrehumbert, & Lichtenberg, 2025) and JANUS (Graham et al., 2021, 2022). Radiative transfer: SOCRATES (Manners, 2024; Manners et al., 2017), with measured surface reflection properties (Hammond et al., 2025). Chemistry: VULCAN kinetics (Tsai et al., 2017, 2021) and FastChem equilibrium (Kitzmann et al., 2024). Outgassing: CALLIOPE (Bower et al., 2022; Nicholls et al., 2024; Shorttle et al., 2024) and atmodeller (Bower et al., 2025), with LavAtmos rock vapour (van Buchem et al., 2023). Escape: ZEPHYRUS (Postolec, Lichtenberg, Nicholls, et al., 2026). Star: MORS (Johnstone et al., 2021). Tides and orbit: LovePy (Hay & Matsuyama, 2019; Nicholls, Guimond, et al., 2025) and Obliqua (van Dijk et al., 2026). Accretion: Morrigan (Kimura et al., 2025). Observation and inference: petitRADTRANS (Mollière et al., 2019; Nasedkin et al., 2024) and a Bayesian optimisation module (Nicholls, Lichtenberg, Riegler, et al., 2026) operate on the output and are not shown.

Laboratory data enter PROTEUS as versioned external tables rather than as code, including equations of state, melting curves, gas opacities, aerosol optical properties, and stellar spectra. The tables are downloaded from Zenodo and Dataverse and verified on first use, so a new measurement replaces a table and leaves the code unchanged. A new module implements the interface of its role and is registered in the configuration schema. A single validated TOML file defines each simulation, which makes every published run reproducible.

Each pull request to PROTEUS or a module runs a fast set of unit tests with the physics mocked, and coverage must stay above fixed thresholds. A nightly suite adds runs of the actual solvers and the coupled integration tests. Every new test has to include an edge case, and physics tests also have to check a physical invariant such as conservation of mass or energy, positivity, or agreement with a reference value. The modules are also validated against analytic limits and against each other where two implementations of one role exist, such as the magma ocean energy budget in Aragog and in SPIDER (Lichtenberg, Attia, et al., 2026). The interior structure is validated against other published models up to 20 Earth masses (Lichtenberg, Attia, et al., 2026). The coupled evolution is benchmarked against the other codes in the CHILI intercomparison (Lichtenberg, Schaefer, et al., 2026; Nicholls, Krissansen-Totton, et al., 2026). Finally, the framework is tested against exoplanet observations (Nicholls, Lichtenberg,

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Chatterjee, et al., 2026; Postolec, Lichtenberg, Teske, et al., 2026; Sastre, Lichtenberg, Dang, et al., 2026) and the Hadean geological record (van Dijk et al., 2026). Test counts and coverage of every module are published on the validation page, and the documentation covers installation, configuration, and tutorials.

Research impact statement

PROTEUS opened evolutionary models to geochemistries far from the oxidised mantles usually assumed for Earth, Venus, and Mars (Lichtenberg et al., 2021; Nicholls et al., 2024). Its simulations have since revealed phenomena that arise only when planetary subsystems are coupled, with consequences for Solar System science, exoplanet astronomy, and astrobiology.

In Solar System science, PROTEUS connects the magma ocean epoch of the terrestrial planets to their geological record. On the Hadean Earth, a feedback between the tides of the young Moon and greenhouse forcing may have set the onset of habitable surface conditions (van Dijk et al., 2026).

In exoplanet astronomy, the simulations indicate that the interiors of many low-mass planets stay molten. A self-limiting feedback between radiative cooling, tidal heating, and mantle rheology prolongs magma ocean epochs (Nicholls, Guimond, et al., 2025). Above a magma ocean, convection can shut down and leave deep radiative layers in the atmosphere (Nicholls, Pierrehumbert, Lichtenberg, Soucasse, et al., 2025). Most rocky sub-Neptunes are expected to retain molten interiors (Calder et al., 2026), whereas the interior of a super-Earth contracts by about 10 per cent as it crystallises (Lichtenberg, Attia, et al., 2026). PROTEUS provides predictions for the JWST lava-planet survey of Sastre, Lichtenberg, Dang, et al. (2026). The simulations also offer an evolutionary interpretation of JWST observations of TOI-561 b (Postolec, Lichtenberg, Teske, et al., 2026) and L 98-59 d (Nicholls, Lichtenberg, Chatterjee, et al., 2026), consistent with a present-day magma ocean on both planets.

For astrobiology, the simulations show which rocky planets can keep their volatiles at the surface. On young planets, the balance between outgassing and escape determines whether a bare rock or a water- or carbon-rich atmosphere remains (Postolec, Lichtenberg, Nicholls, et al., 2026). Late in their evolution, the atmospheres of highly irradiated super-Earths can re-inflate through redox-driven processes (Cesario et al., 2026), and sulfur species provide an observational tracer of the mantle redox state (Panagiotou et al., 2026).

More than 20 contributors have developed the framework openly on GitHub since 2018. Releases of the framework and of its Python modules are tagged, published on PyPI, and archived with a DOI on Zenodo.

AI usage disclosure

Generative AI tools have assisted the development of PROTEUS since early 2026, and were used in the preparation of this paper. We mainly use Claude Code (Anthropic; Fable, Opus, and Sonnet model families) to draft and refactor code, scaffold tests, and revise and extend documentation. Human authors steer, review, edit, and validate all AI-assisted output, make all design decisions, and are responsible for the content of the software and of this paper.

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